COMP 3541: Project Proposal <REVISED>
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As per the final project instructions, I have included my revisions to this proposal, as they apply to my final project.

PROJECT OVERVIEW

For my final project I am wanting to build a job tracking application that I would use to keep track of the jobs which I am looking to apply to following graduation. I feel this is very relevant and useful for myself, but will also provide a great opportunity to implement the requirements of this assignment.

PROJECT UI GENERAL NAVIGATION <All Implemented>

A. The user will be greeted with a login page, where they will enter their user credentials.
B. After entering valid user credentials, the user will see a page which lists the jobs that they have entered into the database. These jobs are listed based on the priority which the user selected in a previous session. This priority listing can be modified at any time by the user.
    a. The jobs that each user enters and sees will be specific to that user and are not globally available to all users.
C. There will be a "delete" button beside each item in the list for the user to delete any entries they no longer want in the list.
D. In the website header, there will be a search bar for the user to search their job list for specific terms to isolate jobs of interest.
E. On the screen there will also be an obvious "add job" button which, when clicked, takes the user to a form where they can enter all the information for a new job that they have found. Once added, this job will be viewable in the job list page.
F. When the user is done with their session, they can click the "log out" button. All of their data is saved within the database.
G. Also included an update job page as well (the "Select" button).

PROJECT DATABASE SCHEMA:

There will be 4 tables in my database. I will briefly outline these below with sample values included. Please also see my relational ERD draft (bottom of this document) for an overview of the attribute data-types and table relationships.

1. **USER TABLE**
    a. The user table stores all the information for the user.
    b. The userID is used to keep entries in all other tables unique to the user for queries. This allows the user to personalize their job application list and organize it in a manner that works for them. It also aids with querying the database for their entries. NOTE: <PK> = primary key, <FK> = foreign key.

TABLE 1: USER

User ID <PK>	First Name	Last Name	Username	Password
U001	Jane	Doe	JD0e	<hashed password>

2. **COMPANY TABLE**
    a. Stores the general information for specific companies.
    b. These companies can then be selected and reused in form entries.
    c. Removed Type and Location from this table as it seemed non-beneficial to the application.

TABLE 2: COMPANY				
Company ID <PK>	User ID <FK>	Company name	~~Company type~~	~~Company Location~~
C001	U001	Suunto	~~Athletics~~	~~USA, Canada~~

3. **JOB TABLE**
    a. The job table stores general details for each job description.
    b. This is to allow the user to try to organize jobs by similar job names, so the job titles can be reused between entries. As I find this is a tricky part of job applications is knowing which job titles to search for. I feel this structure will aid with this organization.
    c. Removed Job Description as it seemed unnecessary given the tasks and qualifications give all relevant info for the user.

TABLE 3: JOB			
Job ID <PK>	User ID <FK>	Job Title	~~Job Description~~
J001	U001	Algorithm Software Engineer	~~Designing algorithms for the following.....~~

4. **COMPANY_JOB TABLE**
    a. This is a bridge table between the COMPANY and JOB tables to avoid the many-many relationship that these tables would have otherwise.
    b. This table stores the more specific details of specific jobs as they pertain to the company and user in question.
    c. I added a "Job Image" in order to practice utilizing files and images in the application.

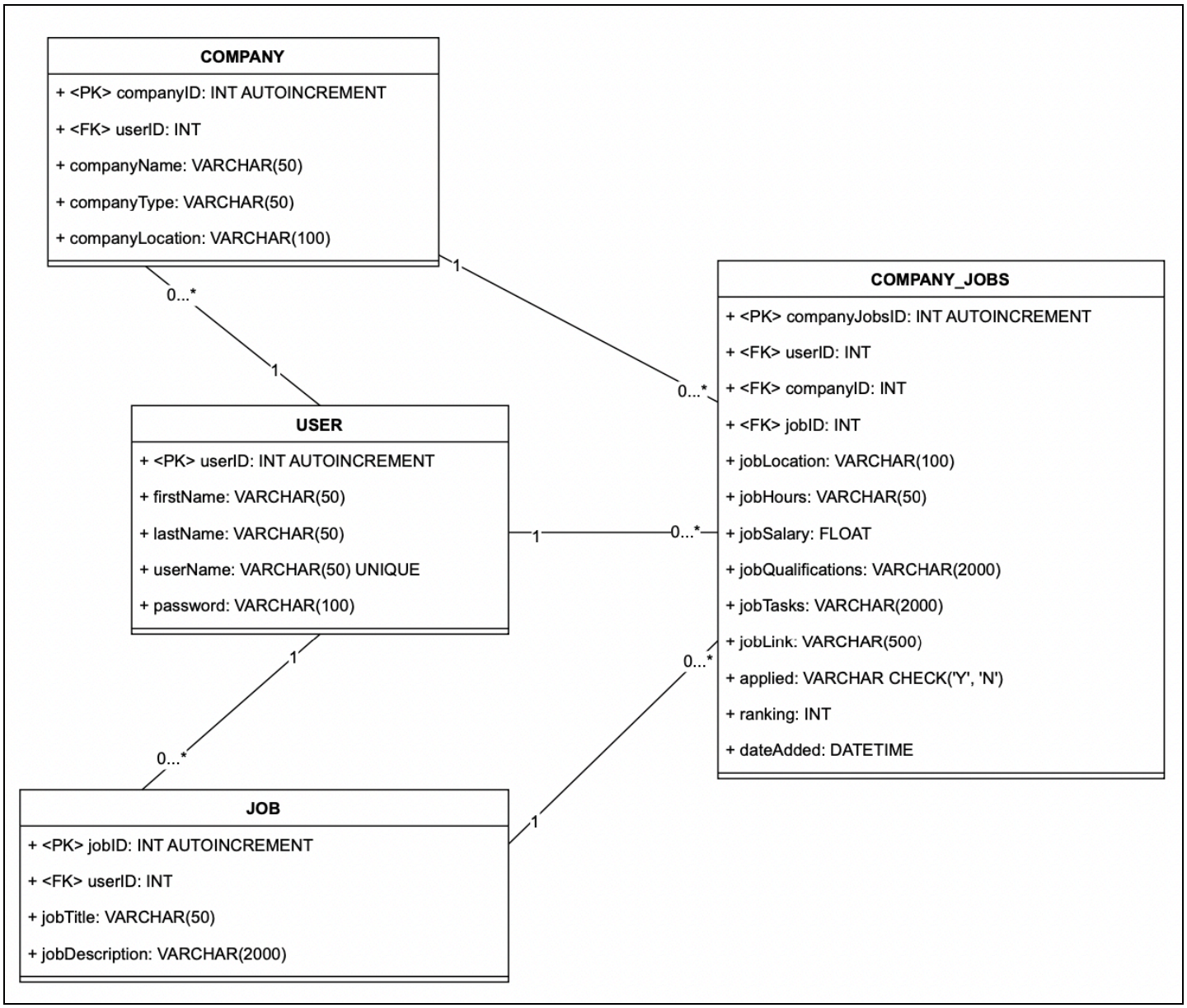
TABLE 4: COMPANY_JOBS						
Company_jobs ID <PK>	**User ID <FK>**	**Company ID <FK>**	**Job ID <FK>**	**Job Location**	**Job Hours**	**Job Salary**
CJ001	U001	C001	J001	Vancouver, BC	Full-time	$96,000

Main Job qualifications	**Essential job tasks**	**Job link**	**applied**	**Ranking (priority)**	**Date added**	**Job Image**
Requires a Bachelor's degree in	Code in C, C++.....	URL here	Yes	2	August 2, 2025	../images/image.jpg

DESIGN RATIONALE:<Implemented all except password hashing>

With this design, I will have multiple, dynamic web pages which allow for database interaction with data storage, retrieval, updating, deletion, ordering and searching data. With user specific sessions, I can implement user authentication, access control, and user specific views. I can practice applying privacy and security standards ~~such as securely hashing and storing a user password~~. I plan to apply AJAX and advanced PHP features for efficient and convenient user experience. I also feel it will provide a great opportunity to apply the MVC design model within my database schema. Altogether this will aid in creating a modular design that is organized and can be modified as needed. Please let me know if you have any concerns or further questions about my design idea. Thank you!

DESIGN ERD: <See Updated UML attached to Final Project Submission>


COMPANY
+ <PK> companyID: INT AUTOINCREMENT
+ <FK> userID: INT
+ companyName: VARCHAR(50)
+ companyType: VARCHAR(50)
+ companyLocation: VARCHAR(100)
COMPANY_JOBS
+ <PK> companyJobsID: INT AUTOINCREMENT
+ <FK> userID: INT
+ <FK> companyID: INT
+ <FK> jobID: INT
+ jobLocation: VARCHAR(100)
+ jobHours: VARCHAR(50)
+ jobSalary: FLOAT
+ jobQualifications: VARCHAR(2000)
+ jobTasks: VARCHAR(2000)
+ jobLink: VARCHAR(500)
+ applied: VARCHAR CHECK('Y', 'N')
+ ranking: INT
+ dateAdded: DATETIME
USER
+ <PK> userID: INT AUTOINCREMENT
+ firstName: VARCHAR(50)
+ lastName: VARCHAR(50)
+ userName: VARCHAR(50) UNIQUE
+ password: VARCHAR(100)
JOB
+ <PK> jobID: INT AUTOINCREMENT
+ <FK> userID: INT
+ jobTitle: VARCHAR(50)
+ jobDescription: VARCHAR(2000)
1
0...*
1
0...*
1
0...*
1
0...*
1
0...*